

# Displays

D1:

$$R_{[t_0, t_1]} = \{\Phi(t_1; t_0, x_0, d) : x_0 \in \mathcal{X}_0, d \in \mathcal{D}\}. \quad (2.1)$$

D2:

$$P_X^N(\{x_1, \dots, x_N : P_X(c(\epsilon, \delta; x_1, \dots, x_N)) \geq 1 - \epsilon\}) \geq 1 - \delta. \quad (2.2)$$

D3:

$$\hat{\kappa}^{-1}(x) = z_m(x)^\top \hat{M}_{m,\sigma}^{-1} z_m(x) \quad (2.3)$$

D4:

$$\hat{M}_{m,\sigma} = (\sigma^2 I + \frac{1}{N} Z Z^\top)^{-1} = \sigma^{-2} \left( I - Z (\sigma^2 N I + Z^\top Z)^{-1} Z^\top \right). \quad (2.4)$$

D5:

$$\hat{\kappa}^{-1}(x) = N \sigma_0^{-2} z_m(x)^\top z_m(x) - N \sigma_0^{-2} z_m(x)^\top Z (\sigma_0^2 I + Z^\top Z)^{-1} Z^\top z_m(x), \quad (2.5)$$

D6:

$$\kappa^{-1}(x) = k(x, x) - k_D(x)^\top (\sigma_0^2 I + K)^{-1} k_D(x), \quad (2.6)$$

D7:

$$K_{ij} = k(x_i, x_j), \quad (k_D(x))_i = k(x_i, x). \quad (2.7)$$

D8:

$$N \geq \frac{5}{\epsilon} \left( \log \frac{4}{\delta} + d \log \frac{40}{\epsilon} \right), \quad (3.1)$$

D9:

$$P_X^N \left( \left\{ x_1, \dots, x_N : D_{\text{ber}}(\hat{r}_Q || r_Q) \leq \frac{D_{KL}(W_Q || W_P) + \log \frac{N+1}{\delta}}{N} \right\} \right) \geq 1 - \delta. \quad (3.2)$$

D10:

$$\begin{aligned} \mathbb{P}(\forall i \geq 1, P_X(\{x : C^i(x) \leq \eta\}) \geq 1 - \epsilon^i) \\ \geq 1 - \delta. \end{aligned}$$

D11:

$$C_P = \{x : g_p(x)^2 \leq \eta\}, \quad C_Q = \{x : g_q(x)^2 \leq \eta\}, \quad (3.3)$$

D12:

$$\text{Var}_{g_q}(x) = k(x, x) - k(X, x)^\top (\sigma^2 I_N + K(X, X))^{-1} k(X, x). \quad (3.4)$$

D13:

$$\bar{c}_Q = \{x : k(x, x) - k(X, x)^\top (\sigma^2 I_N + K(X, X))^{-1} k(X, x) \leq \eta\} = \{x : \kappa^{-1}(x) \leq \eta\} \quad (3.5)$$

D14:

$$\hat{r}_Q = \frac{1}{N} \sum_{i=1}^N 1 - F_1 \left( \frac{\eta}{\kappa^{-1}(x_i)} \right), \quad (3.6)$$

D15:

$$F_1(x) = \mathbb{P}(Z^2 \leq x) \text{ where } Z \sim \mathcal{N}(0, 1). \quad (3.7)$$

D16:

$$\bar{r} = \sup \left\{ \beta : D_{\text{ber}}(\hat{r}_Q || \beta) \leq \frac{D_{KL}(\mathcal{N}(0, (K^{-1} + \sigma_0^{-2} I)^{-1}) || \mathcal{N}(0, K)) + \log \frac{N+1}{\delta}}{N} \right\}, \quad (3.8)$$

D17:

$$\begin{aligned} C_P &= \{x : (W_P^\top z_m(x))^2 \leq \eta\}, \\ C_Q &= \{x : (W_Q^\top z_m(x))^2 \leq \eta\}, \end{aligned} \quad (3.9)$$

D18:

$$\bar{c}_Q = \{x : \mathbb{E}[(W_Q^\top z_m(x))^2] \leq \eta\} = \{x : z_m(x)^\top \hat{M}_{m,\sigma_0}^{-1} z_m(x) \leq \eta\}, \quad (3.10)$$

D19:

$$\bar{r} = \sup \left\{ \beta : D_{\text{ber}}(\hat{r}_Q || \beta) \leq \frac{D_{KL}(\mathcal{N}(0, (\sigma_0^2 I + \hat{M}_{m,\sigma_0})^{-1}) || \mathcal{N}(0, \sigma_0^{-2} I)) + \log \frac{N+1}{\delta}}{N} \right\}. \quad (3.11)$$

D20:

$$\tilde{K} = K_{Nr} K_{rr}^{-1} K_{Nr}, \quad (3.12)$$

D21:

$$\begin{aligned} \tilde{\kappa}^{-1}(x) = & k(x, x) - \sigma_0^{-2} \left( k_D(x)^\top k_D(x) \right. \\ & \left. - (K_{Nr} k_D(x))^\top (I - K_{Nr} (\sigma_0^2 K_{rr} + K_{rN} K_{Nr})^{-1} (K_{rN} k_D(x))) \right). \end{aligned} \quad (3.13)$$

D22:

$$D_{KL}(Z_0 || Z_1) = \frac{1}{2} \log \det \Sigma_1 \Sigma_0^{-1} + \frac{1}{2} \text{tr} \Sigma_1^{-1} ((\mu_0 - \mu_1)(\mu_0 - \mu_1)^\top + \Sigma_0) - \frac{N}{2}. \quad (3.14)$$

D23:

$$\frac{1}{2} \log \det (I + \sigma_0^{-2} K) + \frac{1}{2} \text{tr} ((I + \sigma_0^{-2} K)^{-1}) - \frac{N}{2}. \quad (3.15)$$

D24:

$$= \frac{1}{2} \sum_{i=1}^N \left( \log(1 + \sigma_0^{-2} \lambda_i) + \frac{1}{1 + \sigma_0^{-2} \lambda_i} - 1 \right). \quad (3.16)$$

D25:

$$\begin{aligned} \dot{x}_1 &= \frac{1}{T} (d - \min(c, vx_1, w(\bar{x} - x_2))) \\ \dot{x}_i &= \frac{1}{T} (\min(c, vx_{i-1}, w(\bar{x} - x_i)) \\ &\quad - \min(c, vx_i, w(\bar{x} - x_{i+1}))), \quad (i = 2, \dots, n-1) \\ \dot{x}_n &= \frac{1}{T} (\min(c, vx_{n-1}, w(\bar{x} - x_n)/\beta) - \min(c, vx_n)), \end{aligned} \quad (4.1)$$

D26:

$$m_q(x) = k_D(x)^\top (\sigma^2 I_N + K)^{-1} h, \quad (A.1)$$

D27:

$$k_q(x, y) = k(x, y) - k_D(x)^\top (\sigma^2 I_N + K)^{-1} k_D(y). \quad (A.2)$$

D28:

$$m_q(x) = \sigma^{-2} b(x)^\top (\Sigma^{-1} + \sigma^2 B B^\top)^{-1} B y \quad (A.3)$$

D29:

$$k_q(x, y) = b(x)^\top (\Sigma^{-1} + \sigma^{-2} B B^\top)^{-1} b(y), \quad (A.4)$$

D30:

$$\begin{aligned} \mathbb{E}[\ell(C_Q, x)] &= \mathbb{E}[\mathbb{1}\{x \in C_Q\}] \\ &= 1 - \mathbb{P}(g_p(x)^2 \leq \eta) = 1 - \mathbb{P}\left(\frac{g_p(x)^2}{\kappa^{-1}(x)} \leq \frac{\eta}{\kappa^{-1}(x)}\right) \\ &= 1 - F_1\left(\frac{\eta}{\kappa^{-1}(x)}\right). \end{aligned} \quad (B.1)$$

D31:

$$P_X^N(\{x_1, \dots, x_N : D_{\text{ber}}(\hat{r}_Q || r_Q) \leq \gamma\}) \geq 1 - \delta, \quad (B.2)$$

D32:

$$\begin{aligned} K_q(X, X) &= K(X, X) - K(X, X) (\sigma_0^2 I + K(X, X))^{-1} K(X, X) \\ &= (K(X, X)^{-1} + \sigma_0^{-2} I)^{-1}. \end{aligned} \quad (B.3)$$

D33:

$$\mathbb{P}((g(x))^2 \geq \eta) \leq \mathbb{P}((g(x))^2 \geq \mathbb{E}[(g(x))^2]) = \mathbb{P}\left(\frac{(g(x))^2}{\mathbb{E}[(g(x))^2]} > 1\right) = 1 - F_1(1). \quad (B.4)$$

D34:

$$\mathbb{P}((g(X))^2 > \eta) = \int_{\mathcal{X}} \mathbb{P}((g(x))^2 > \eta) dP_x(x) \quad (B.5)$$

D35:

$$= \int_{\mathcal{X}} \mathbb{P}((g(x))^2 > \eta) \mathbb{1}\{\mathbb{E}[(g(x))^2] > \eta\} dP_x(x) \quad (B.6)$$

D36:

$$+ \int_{\mathcal{X}} \mathbb{P}((g(x)^2 > \eta) \nmid \{\mathbb{E}[(g(x)^2) \leq \eta]\} dP_x(x). \quad (\text{B.7})$$

D37:

$$\mathbb{P}((g(X)^2 > \eta) = \int_{\mathcal{X}} \mathbb{P}((g(x)^2 > \eta) dP_x(x) \quad (\text{B.8})$$

D38:

$$\geq \int_{\mathcal{X}} \mathbb{P}((g(x)^2 > \eta) \nmid \{\mathbb{E}[(g(x)^2) > \eta]\} dP_x(x) \quad (\text{B.9})$$

D39:

$$\geq (1 - F_1(1)) \int_{\mathcal{X}} \mathbb{P}(\mathbb{E}[(g(x)^2) > \eta] dP_x(x) \quad (\text{B.10})$$

D40:

$$= (1 - F_1(1)) \mathbb{P}(\mathbb{E}[(g(x)^2) > \eta) = (1 - F_1(1))r(\hat{c}_\eta), \quad (\text{B.11})$$

## Inline

I1:  $\mathbb{R}^n$

I2:  $\Phi(t_1; t_0, x_0, d)$

I3:  $x_0$

I4:  $t_0$

I5:  $d$

I6:  $t_1$

I7:  $\mathcal{X}_0$

I8:  $\mathcal{D}$

I9:  $t_0, t_1$

I10:  $R_{[t_0, t_1]}$

I11:  $\hat{R}_{[t_0, t_1]}$

I12:  $\mathbb{E}[\cdot]$

I13:  $\mathbb{P}(\cdot)$

I14:  $D_{KL}(P||Q)$

I15:  $P$

I16:  $Q$

I17:  $D_{ber}(p||q)$

I18:  $p$

I19:  $q$

I20:  $X$

I21:  $F_1$

I22:  $\mathcal{X}$

I23:  $P_X$

I24:  $P_X^N$

I25:  $N$

I26:  $\epsilon$

I27:  $\delta$

I28:  $\mathcal{C}$

I29:  $\bar{c}_Q$

I30:  $W_P$

I31:  $W_Q$

I32:  $C_P$

I33:  $C_Q$

I34:  $\ell(c, x)$

I35:  $c$

I36:  $x$

I37:  $r(c)$

I38:  $x \sim X$

I39:  $\hat{r}(c)$   
 I40:  $x_1, \dots, x_N$   
 I41:  $r_Q$   
 I42:  $c \sim Q$   
 I43:  $\hat{r}_Q$   
 I44:  $M_m$   
 I45:  $\hat{M}_m$   
 I46:  $\leq m$   
 I47:  $\bar{M}_{m, \sigma_0}$   
 I48:  $\sigma_0$   
 I49:  $z_m(x)$   
 I50:  $\hat{\kappa}^{-1}(x)$   
 I51:  $C(x)$   
 I52:  $m$   
 I53:  $k$   
 I54:  $m_q$   
 I55:  $k_q$   
 I56:  $K$   
 I57:  $K_{ij} = k(x_i, x_j)$   
 I58:  $k_D$   
 I59:  $(k_D(x))_i = k(x_i, x)$   
 I60:  $\mathcal{N}(\mu, \Sigma)$   
 I61:  $\mu$   
 I62:  $\Sigma$   
 I63:  $\mathcal{GP}(m, k)$   
 I64:  $\Phi(t_1; t_0, x_0, d)$   
 I65:  $x(t_0) = x_0 \in \mathbb{R}^n$   
 I66:  $d : [t_0, t_1] \rightarrow \mathbb{R}^w$   
 I67:  $\dot{x}(t) = f(t, x(t), d(t))$   
 I68:  $[t_0, t_1]$   
 I69:  $x(t_0) = x_0$   
 I70:  $\mathcal{X}_0 \subset \mathbb{R}^n$   
 I71:  $X_0$   
 I72:  $D$   
 I73:  $X = \Phi(t_1; t_0, X_0, D)$   
 I74:  $P_X(A)$   
 I75:  $A$   
 I76:  $P_X(A) = 1$   
 I77:  $R_{[t_0, t_1]} \subseteq A$   
 I78:  $A \subseteq \mathbb{R}^n$   
 I79:  $B \subseteq \mathbb{R}^n$   
 I80:  $B$   
 I81:  $P_X(\hat{R}_{[t_0, t_1]})$   
 I82:  $\hat{R}_{[t_0, t_1]} = \mathbb{R}^n$   
 I83:  $p_X$   
 I84:  $\{x : p_X(x) \neq 0\}$   
 I85:  $\epsilon, \delta \in (0, 1)$   
 I86:  $\mathcal{X} \subseteq \mathbb{R}^n$   
 I87:  $x_1, \dots, x_N \stackrel{\text{i.i.d.}}{\sim} X$   
 I88:  $c(\epsilon, \delta; x_1, \dots, x_N) \subset \mathcal{X}$   
 I89:  $P_X(c(\epsilon, \delta; x_1, \dots, x_N)) \geq 1 - \epsilon$   
 I90:  $1 - \epsilon$   
 I91:  $1 - \delta$   
 I92:  $P_X^N(A)$   
 I93:  $A \subseteq \mathcal{X}^N$   
 I94:  $x_1, \dots, x_N \in A$

I95:  $P_X^N(\{x_1, \dots, x_N : \dots\}) \geq 1 - \delta$   
 I96:  $c(\epsilon, \delta; x_1, \dots, x_N)$   
 I97:  $\kappa(x) = 1/z_m(x)^\top M_m^{-1} z_m(x)$   
 I98:  $M_m$   
 I99:  $M_m = \int_{\mathcal{X}} z_m(x) z_m(x)^\top dP_X(x)$   
 I100:  $M_m^{-1}$   
 I101:  $\kappa(x)^{-1} = z_m(x)^\top M_m^{-1} z_m(x)$   
 I102:  $2m$   
 I103:  $\hat{M}_m = \frac{1}{N} \sum_{i=1}^N z_m(x_i) z_m(x_i)^\top$   
 I104:  $N \geq \binom{n+m}{n}$   
 I105:  $\hat{M}_{m,\sigma} = \sigma^2 I + \frac{1}{N} \sum_{i=1}^N z_m(x_i) z_m(x_i)^\top$   
 I106:  $\sigma^2 > 0$   
 I107:  $\sigma^2 I$   
 I108:  $\hat{M}_{m,\sigma}$   
 I109:  $\frac{1}{N} \sum_{i=1}^N z_m(x_i) z_m(x_i)^\top$   
 I110:  $\frac{1}{N} Z Z^\top$   
 I111:  $Z \in \mathbb{R}^{\binom{n+m}{n} \times N}$   
 I112:  $Z = [z_m(x_1) \quad \dots \quad z_m(x_N)]$   
 I113:  $\hat{M}_{m\sigma}$   
 I114:  $\sigma^2 = \sigma_0^2/N$   
 I115:  $z_m$   
 I116:  $z_m(x_i)^\top z_m(x_j)$   
 I117:  $Z^\top Z \in \mathbb{R}^{N \times N}$   
 I118:  $(Z^\top Z)_{ij} = z_m(x_i)^\top z_m(x_j)$   
 I119:  $Z^\top z_m(x)$   
 I120:  $(Z^\top z_m(x))_i = z_m(x_i)^\top z_m(x)$   
 I121:  $1$   
 I122:  $k : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$   
 I123:  $\sigma_0^2/N$   
 I124:  $K \in \mathbb{R}^{N \times N}$   
 I125:  $k_D(x) \in \mathbb{R}^N$   
 I126:  $(K)_{ij} = k(x_i, x_j)$   
 I127:  $m \in \mathbb{N}_+$   
 I128:  $\sigma_0^2 \in \mathbb{R}_{++}$   
 I129:  $N \leftarrow \lceil \frac{5}{\epsilon} \left( \log \frac{4}{\delta} + \binom{n+2m}{n} \log \frac{40}{\epsilon} \right) \rceil$   
 I130:  $i \in \{1, \dots, N\}$   
 I131:  $x_i \sim X$   
 I132:  $\hat{M}_{m,\sigma_0} \leftarrow \sigma_0^2 I + \frac{1}{N} \sum_{i=1}^N z_m(x_i) z_m(x_i)^\top$   
 I133:  $\alpha \leftarrow \max_i z_m(x_i)^\top \hat{M}_{m,\sigma_0}^{-1} z_m(x_i)$   
 I134:  $C(x) = z_m(x)^\top \hat{M}_{m,\sigma_0}^{-1} z_m(x)$   
 I135:  $\not\models \{C(x) \leq \alpha\}$   
 I136:  $(x_1, \dots, x_n) \in \mathbb{R}^n$   
 I137:  $x_1, \dots, x_s$   
 I138:  $s < n$   
 I139:  $s$   
 I140:  $\mathbb{R}^s$   
 I141:  $c \subseteq \mathcal{X}$   
 I142:  $\mathcal{C} \subseteq 2^{\mathcal{X}}$   
 I143:  $r(c) = \mathbb{E}[\ell(c, X)]$   
 I144:  $\ell : \mathcal{C} \times \mathcal{X} \rightarrow \mathbb{R}_+$   
 I145:  $\hat{r}(c) = \frac{1}{N} \sum_{i=1}^N \ell(c, x_i)$   
 I146:  $r(c) - \hat{r}(c)$   
 I147:  $\leq d$   
 I148:  $\ell : \mathcal{C} \times \mathcal{X} \rightarrow \{0, 1\}$

I149:  $\{0, 1\}$   
I150:  $\hat{r}(c) = 0$   
I151:  $P_X^N(\{x_1, \dots, x_N : r(c) \leq \epsilon\}) \geq 1 - \delta$   
I152:  $2k$   
I153:  $V$   
I154:  $g : \mathbb{R}^n \rightarrow \mathbb{R}$   
I155:  $\text{Pos}(V) = \{ \{x : g(x) \geq 0\}, g \in V \}$   
I156:  $\{x \in \mathcal{X} : C(x) \leq \alpha\}$   
I157:  $\alpha = \max_i C(x_i)$   
I158:  $P_X^N(\{x_1, \dots, x_N : P_X(\{x \in \mathcal{X} : C(x) \leq \alpha\}) \geq 1 - \epsilon\}) \geq 1 - \delta$   
I159:  $\epsilon, \delta$   
I160:  $\mathcal{C} = \text{Pos}(\mathbb{R}[x]_{2m}^n)$   
I161:  $\ell(c, x) = \mathbb{1}_{\{x \notin c\}}$   
I162:  $\{x \in \mathbb{R}^n : C(x) \leq \alpha\}$   
I163:  $\text{Pos}(\mathbb{R}[x]_{2m}^n)$   
I164:  $c = \{x \in \mathbb{R}^n : \alpha - C(x) \geq 0\}$   
I165:  $\mathbb{R}[x]_{2m}^n$   
I166:  $\binom{n+2m}{n}$   
I167:  $\leq \binom{n+2m}{n} = d$   
I168:  $r(c) = \mathbb{E}[\mathbb{1}_{\{x \notin c\}}] = 1 - P_X(c)$   
I169:  $\hat{r}(c) = \sum_{i=1}^N \mathbb{1}_{\{x_i \notin c\}}$   
I170:  $\hat{r}(\{x \in \mathbb{R}^n : C(x) \leq \alpha\}) = 0$   
I171:  $\ell$   
I172:  $N \geq \frac{5}{\epsilon} (\log \frac{4}{\delta} + \binom{n+2m}{n} \log \frac{40}{\epsilon})$   
I173:  $P_X^N(\{x_1, \dots, x_N : 1 - P_X(\{x \in \mathbb{R}^n : C(x) \leq \alpha\}) \leq \epsilon\}) \geq 1 - \delta$   
I174:  $\square$   
I175:  $r_Q = \mathbb{E}[\ell(c, X)]$   
I176:  $\hat{r}_Q = \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^N \ell(c, x_i)\right]$   
I177:  $w \in \mathcal{W}$   
I178:  $\mathcal{W}$   
I179:  $D_{KL}(W_P || W_Q)$   
I180:  $D_{ber}(q || p)$   
I181:  $D_{ber}(q || p) = q \log \frac{q}{p} + (1 - q) \log \frac{1-q}{1-p}$   
I182:  $N_0$   
I183:  $N_b$   
I184:  $\eta$   
I185:  $N \leftarrow N_0$   
I186:  $D \leftarrow (x_1, \dots, x_N) \stackrel{\text{i.i.d.}}{\sim} X$   
I187:  $i \leftarrow 0$   
I188:  $\epsilon^0 \leftarrow 1$   
I189:  $\epsilon^i > \epsilon$   
I190:  $i \leftarrow i + 1$   
I191:  $(x_{N+1}, \dots, x_{N+N_b}) \stackrel{\text{i.i.d.}}{\sim} X$   
I192:  $N \leftarrow N + N_b$   
I193:  $K_{\sigma_0} \leftarrow \sigma_0^2 I + K$   
I194:  $C : \mathcal{X} \rightarrow \mathbb{R}_+$   
I195:  $C(x) = k(x, x) - k_D(x) K_{\sigma_0}^{-1} k_D(x)$   
I196:  $\bar{r}$   
I197:  $\epsilon_i \leftarrow \frac{\bar{r} + \frac{2}{N} \log(\frac{\pi^2 i^2}{6\delta})}{1 - F_1(1)}$   
I198:  $\mathbb{1}_{\{C(x) \leq \eta\}}$   
I199:  $\bar{c}_Q = \{x : \kappa^{-1}(x) \leq \eta\}$   
I200:  $r(\bar{c}_Q)$   
I201:  $C^i$   
I202:  $i$   
I203:  $\geq 1 - \epsilon$

I204:  $D_{KL}(\mathcal{N}(0, (\sigma_0^{-1}I + K^{-1})^{-1}) || \mathcal{N}(0, K))$   
 I205:  $o(N)$   
 I206:  $g_p$   
 I207:  $g_q$   
 I208:  $y_1 = \dots = y_N = 0$   
 I209:  $\sigma_0^2$   
 I210:  $m_q = 0$   
 I211:  $\bar{c}_Q = \{x : \mathbb{E}[g_q(x)^2] \leq \eta\}$   
 I212:  $\mathbb{E}[g_q(x)] = m_q(x) = 0$   
 I213:  $x \in \mathcal{X}$   
 I214:  $\mathbb{E}[g_q(x)^2] = \text{Var}_{g_q}(x)$   
 I215:  $C_Q$   
 I216:  $C_P$   
 I217:  $\bar{r} \in (0, 1)$   
 I218:  $(q, p)$   
 I219:  $q = p$   
 I220:  $D_{\text{ber}}(\hat{r}_Q || \beta) - (D_{KL}(\mathcal{N}(0, (K^{-1} + \sigma_0^{-2}I)^{-1}) || \mathcal{N}(0, K)) + \log \frac{N+1}{\delta})/N = 0$   
 I221:  $\beta \in [\hat{r}_Q, 1)$   
 I222:  $r(\bar{c}_\eta)$   
 I223:  $r(\bar{c}_Q) \leq \frac{1}{1-F_1(1)}r_Q \approx 3.15r_Q$   
 I224:  $\epsilon^0 \leftarrow 1$   
 I225:  $i > 0$   
 I226:  $C_Q^i$   
 I227:  $\{g_Q^i(x)^2 \leq \eta\}$   
 I228:  $g_Q^i(x) \sim \mathcal{N}(0, k(x, x) - k_{D^i}(x)^\top (\sigma_0^2 I + K^i) k_{D^i}(x))$   
 I229:  $r_Q^i$   
 I230:  $\forall i \geq 1, \mathbb{P}(r_Q^i > (1 - F_1(1))\epsilon^i) \leq \frac{6\delta}{\pi^2 i^2}$   
 I231:  $\mathbb{P}(\exists i, r_Q^i > (1 - F_1(1))\epsilon^i) \leq \sum_{i \geq 1} \frac{6\delta}{\pi^2 i^2} = \delta$   
 I232:  $r_Q^i \leq \epsilon^i$   
 I233:  $\forall i \geq 1, P_X(\{x : C^i(x) > \eta\}) \leq \frac{r_Q^i}{1-F_1(1)} = \epsilon^i$   
 I234:  $k(x, y) = z_m(x)^\top z_m(y)$   
 I235:  $N \times N$   
 I236:  $W_P \sim \mathcal{N}(0, \sigma_0^{-2}I)$   
 I237:  $W_Q \sim \mathcal{N}(0, \hat{M}_{m, \sigma_0}^{-1})$   
 I238:  $W_P^\top z_m$   
 I239:  $W_Q^\top z_m$   
 I240:  $\bar{c}_Q$   
 I241:  $k(x, y) = \exp(-\|x - y\|^2 / (2\ell)^2)$   
 I242:  $[0, 1]$   
 I243:  $0$   
 I244:  $1$   
 I245:  $\eta = \binom{n+2m}{n} / \epsilon$   
 I246:  $\binom{n+2m}{n} / \epsilon$   
 I247:  $\kappa^{-1}(x)$   
 I248:  $O(N^3)$   
 I249:  $r$   
 I250:  $K_{N^r} \in \mathbb{R}^{N \times r}$   
 I251:  $K_{rr} \in \mathbb{R}^{r \times r}$   
 I252:  $i, j$   
 I253:  $k(x_i, x_j)$   
 I254:  $K \mapsto \tilde{K}$   
 I255:  $r \times r$   
 I256:  $D_{KL}(Z_0 || Z_1)$   
 I257:  $Z_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$

I258:  $Z_1 \sim \mathcal{N}(\mu_1, \Sigma_1)$   
 I259:  $\Sigma_0 = (\sigma_0^{-2}I + K^{-1})^{-1}$   
 I260:  $\Sigma_1 = K$   
 I261:  $\mu_0 = \mu_1 = 0$   
 I262:  $\log(1 + \sigma_0^{-2}x)$   
 I263:  $1/(1 + \sigma_0^{-2}x)$   
 I264:  $x \geq 0$   
 I265:  $\lambda_1, \dots, \lambda_N$   
 I266:  $O(N^2)$   
 I267:  $\lambda_p$   
 I268:  $p^{th}$   
 I269:  $\lambda_i$   
 I270:  $\lambda_i \approx \lambda_p$   
 I271:  $\lambda_i < \lambda_p$   
 I272:  $\epsilon = 0.1$   
 I273:  $\delta = 10^{-9}$   
 I274:  $\eta = 0.15$   
 I275:  $\dot{z} = y, \dot{y} = -\alpha y + z - z^3 + \gamma \cos(\omega t)$   
 I276:  $x = (z, y) \in \mathbb{R}^2$   
 I277:  $\alpha, \gamma, \omega \in \mathbb{R}$   
 I278:  $\alpha$   
 I279:  $\gamma$   
 I280:  $\omega$   
 I281:  $\alpha = 0.05$   
 I282:  $\gamma = 0.4$   
 I283:  $\omega = 1.3$   
 I284:  $z(0) \in [0.95, 1.05]$   
 I285:  $y(0) \in [-0.05, 0.05]$   
 I286:  $[t_0, t_1] = [0, 100]$   
 I287:  $k = 10$   
 I288:  $\epsilon = 0.10$   
 I289:  $k(x, y) = \exp(\|x - y\|^2/(2\ell^2))$   
 I290:  $\ell = 0.25$   
 I291:  $r = 2000$   
 I292:  $\ddot{p}_x = u_1 K \sin(\theta), \ddot{p}_h = -g + u_1 L \cos(\theta), \ddot{\theta} = -d_0 \theta - d_1 \dot{\theta} + n_0 u_2$   
 I293:  $p_x$   
 I294:  $p_h$   
 I295:  $\theta$   
 I296:  $\theta = 0$   
 I297:  $h$   
 I298:  $u_1$   
 I299:  $u_2$   
 I300:  $g = 9.81$   
 I301:  $L = 0.64$   
 I302:  $d_0 = 70$   
 I303:  $d_1 = 17$   
 I304:  $n_0 = 55$   
 I305:  $p_x(0) \in [-1.7, 1.7]$   
 I306:  $\dot{p}_x(0) \in [-0.8, 0.8]$   
 I307:  $p_h(0) \in [0.3, 2.0]$   
 I308:  $\dot{p}_h(0) \in [-1.0, 1.0]$   
 I309:  $\theta(0) \in [-\pi/12, \pi/12]$   
 I310:  $\dot{\theta}(0) \in [-\pi/2, \pi/2]$   
 I311:  $u_1(t) = u_1$   
 I312:  $u_2(t) = u_2$   
 I313:  $\forall t \in [t_0, t_1]$

I314:  $u_1 \in [-1.5 + g/L, 1.5 + g/L], u_2 \in [-\pi/4, \pi/4]$   
 I315:  $[t_0, t_1] = [0, 5]$   
 I316:  $(p_x, p_h)$   
 I317:  $m = 10,000$   
 I318:  $T$   
 I319:  $n$   
 I320:  $x_1, \dots, x_n$   
 I321:  $x_i$   
 I322:  $i^{th}$   
 I323:  $x_1$   
 I324:  $v$   
 I325:  $\bar{x}$   
 I326:  $w$   
 I327:  $u$   
 I328:  $n = 6$   
 I329:  $T = 30$   
 I330:  $v = 0.5$   
 I331:  $w = 1/6$   
 I332:  $\bar{x} = 320$   
 I333:  $x_i(0) \in [100, 200]$   
 I334:  $i = 1, \dots, n$   
 I335:  $d \in [40/T, 60/T]$   
 I336:  $[t_0, t_1] = [0, 4T]$   
 I337:  $x_5$   
 I338:  $x_6$   
 I339:  $(x_5, x_6)$   
 I340:  $g$   
 I341:  $(g(x_1), \dots, g(x_m))$   
 I342:  $m(x) = \mathbb{E}[g(x)]$   
 I343:  $x, y \in \mathcal{X}$   
 I344:  $k(x, y) = \mathbb{E}[g(x)g(y)]$   
 I345:  $b_1, \dots, b_m : \mathcal{X} \rightarrow \mathbb{R}$   
 I346:  $\sum_{i=1}^m w_i b_i$   
 I347:  $w = (w_1, \dots, w_m) \sim \mathcal{N}(0, \Sigma)$   
 I348:  $b_1, \dots, b_m$   
 I349:  $m(x) = 0$   
 I350:  $k(x, y) = \sum_{i=1}^m b(x)^\top \Sigma b(y)$   
 I351:  $b(\cdot) = (b_1(\cdot), \dots, b_m(\cdot))^\top$   
 I352:  $g(x_i) = h_i + \varepsilon$   
 I353:  $\varepsilon$   
 I354:  $\sigma^2$   
 I355:  $B \in \mathbb{R}^m \times N$   
 I356:  $B = [b(x_1) \quad \dots \quad b(x_N)]$   
 I357:  $b = z_k$   
 I358:  $\Sigma = \sigma_0^{-2} I$   
 I359:  $\sigma = N^{-1/2}$   
 I360:  $\text{Var}_{g_q}(x) = z_m(x)^\top \left( \sigma_0^2 I + \frac{1}{N} \sum_{i=1}^N z_m(x_i) z_m(x_i)^\top \right)^{-1} z_m(x)$   
 I361:  $g_p(x) \sim \mathcal{N}(0, \kappa^{-1}(x))$   
 I362:  $g_p(x) / \sqrt{\kappa^{-1}(x)} \sim \mathcal{N}(0, 1)$   
 I363:  $g_p(x)^2 / \kappa^{-1}(x) \sim \chi_1^2$   
 I364:  $g_p(x)^2 / \kappa^{-1}(x)$   
 I365:  $W_P \sim \mathcal{N}(0, \sigma_0^{-2} I)$   
 I366:  $W_Q \sim \mathcal{N}\left(0, (\sigma_0^{-2} I + \hat{M}_{m, \sigma_0})^{-1}\right)$   
 I367:  $W_p$   
 I368:  $W_q$

I369:  $\gamma = \frac{1}{N}(D_{KL}(\mathcal{N}(0, (\sigma_0^2 I + \hat{M}_{m, \sigma_0})^{-1}) || \mathcal{N}(0, \sigma_0^{-2} I)) + \log \frac{N+1}{\delta})$   
 I370:  $D_{\text{ber}}(\hat{r}_Q || r_Q) \leq \gamma$   
 I371:  $\{\beta : D_{\text{ber}}(\hat{r}_Q || \beta) \leq \gamma\}$   
 I372:  $D_{\text{ber}}(\hat{r}_Q || \beta)$   
 I373:  $\beta$   
 I374:  $[0, \infty)$   
 I375:  $\beta = \hat{r}_Q$   
 I376:  $\infty$   
 I377:  $\beta \rightarrow 0$   
 I378:  $\beta \rightarrow 1$   
 I379:  $(0, 1)$   
 I380:  $(g_p(x_1), \dots, g_p(x_N))$   
 I381:  $(g_q(x_1), \dots, g_q(x_N))$   
 I382:  $(x_1, \dots, x_N)$   
 I383:  $K_p(X, X) = K(X, X)$   
 I384:  $\bar{c}_\eta(x) = \mathbb{E}[(g(x)^2)] > \eta$   
 I385:  $W_Q^\top z_m(x)$   
 I386:  $r_Q = \mathbb{P}((g(X)^2 > \eta)$   
 I387:  $\mathbb{P}((g(X)^2 > \eta) \geq \int_{\mathcal{X}} \mathbb{P}((g(x)^2 > \eta) \mathbb{K}\{\mathbb{E}[(g(x)^2)] > \eta\} dP_x(x)$   
 I388:  $r(\bar{c}_\eta) \leq \frac{1}{1-F_1(1)} r_{Q_\eta}$   
 I389:  $\mathbb{P}(\forall i \geq 1, P_X(\{x : C^i(x) \leq \eta\}) \geq 1 - \epsilon^i) \geq 1 - \delta$   
 I390:  $\epsilon^i$   
 I391:  $\epsilon_i$   
 I392:  $W_Q \sim \mathcal{N}(0, \hat{M}_{m, \sigma_0}^{-1})$   
 I393:  $\mathcal{N}(0, (\sigma_0^2 I + \hat{M}_{m, \sigma_0})^{-1})$   
 I394:  $\mathcal{N}(0, (\sigma_0^{-2} I + \hat{M}_{m, \sigma_0})^{-1})$   
 I395:  $(\sigma_0^2 I + K^i)$   
 I396:  $Z = [z_m(x_i) \ \dots \ z_m(x_N)]$   
 I397:  $1/N$   
 I398:  $\sigma_0^{-1} I$   
 I399:  $\sigma^2 I_N$   
 I400:  $D_{KL}(W_Q || W_P)$   
 I401:  $K_{Nr}$   
 I402:  $\exp(\|x - y\|^2 / (2\ell^2))$   
 I403:  $\exp(-\|x - y\|^2 / (2\ell)^2)$   
 I404:  $\sigma^2 B B^\top$   
 I405:  $By$   
 I406:  $b(x)$   
 I407:  $\mathbb{R}m \times N$   
 I408:  $\mathbb{K}\{x \in C_Q\}$   
 I409:  $(g(x))^2$   
 I410:  $r(\hat{c}_\eta)$   
 I411:  $r_{Q_\eta}$   
 I412:  $n = 2$   
 I413:  $N = \lceil \frac{5}{\epsilon} (\log \frac{4}{\delta} + \binom{n+2m}{n} \log \frac{40}{\epsilon}) \rceil$   
 I414:  $m = 10$   
 I415:  $m = 4$